Sanaul Haque

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Synthetic Civilizations as Model Systems for Human and AI Cultural Transmission

Research Proposal

Rayan Sen

Delhi Public School Newtown

Class-XII

India

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Synthetic Civilizations as Model Systems for Human and AI Cultural Transmission

1 Research Topic and Innovation

The proposed study investigates the effects of recurrent transmission of structured cultural knowledge by humans, a big language model, or alternating human and AI transmitters. The use of a synthetic civilization as a model system is its main methodological contribution. A synthetic culture, in contrast to an actual society, may be created from a comprehensive specification, replicated under various experimental settings, and assessed without drawing conclusions about live customs. Mythology, ritual, institutions, causal principles, symbols, and social practices are all intended to be interdependent, making the instrument more than a single narrative. Different types of loss and change are immediately visible thanks to nonce labels, matched fact pairs, contradiction pairs, and embedded canary facts.

This design distinguishes between three processes that are sometimes combined into a broad concept of "drift": introduction of an unsupported claim, modification of a traceable claim, and omission of existing information. Additionally, it checks to see if the destination still has a discernible lineage fingerprint. Instead of being a demonstration based on specific tales, the generated codebook and corpus would serve as a reusable benchmark for researching multi-step human-AI communication.

2 Research Problem and Literature

Information is not replicated impartially, as serial reproduction has long demonstrated. Repeated recounting reduced and conventionalized narratives, according to Bartlett's early tests [1]. Controlled techniques for examining cultural selection and change were created by later transmission-chain research [2]. Transmission bottlenecks can make artificial languages more organized and learnable across generations, as further demonstrated by iterated-learning studies [3]. Experimental claims rely on well selected controls, replication, and clear trade-offs between ecological realism and causal identification, according to recent methodological work [4].

The AI literature reveals related but distinct dynamics. LLM-only chains can amplify small biases and converge toward attractor states whose strength depends on the task and model [5]. While communities of interacting LLMs can create customs and collective biases [7], exposure to AI-generated ideas can change the collective development of human thoughts [6]. While the model-collapse situation is not the same as semantic retelling, recursive training on model-generated data can likewise deteriorate learned distributions [8]. A controlled comparison of human-only, AI-only, and mixed chains conveying the same rich, internally connected culture from a fully known origin is not provided by the research examined here. Therefore, it is still unknown whether AI and humans maintain distinct content classes, regularize inconsistencies at different rates, or become indistinguishable in the event of a severe information bottleneck.

3 Research Question and Hypotheses

The main question is how the fidelity, innovation, handling of contradictions, and structural integrity of a fully specified synthetic culture vary among four generations of human-only, AI-only, and alternating human-AI chains, and how these differences are mitigated by a limited word budget.

H1. Alternating chains will exhibit an intermediate trajectory; AI-only chains will eliminate or resolve intentional inconsistencies before human-only chains.

H2. Fact kind and transmitter lineage will interact with fact survival. While AI chains are predicted to preserve more explicit causal-institutional principles, human chains are expected to retain more sensory-social detail.

H3. Compared to the 500-word condition, the 220-word bottleneck will speed up fact loss and endpoint convergence, with the greatest impact in AI-only and alternating chains.

H4. In order to identify a detectable transmission fingerprint, independent raters will categorize final texts by lineage above the one-third chance threshold.

4 Methodology

Prior to data collection, the confirmatory trial will be preregistered. It is a 3 x 2 design that combines two matching founding cultures with transmission lineage by word restriction. Each cell will run three separate chains for four generations, producing 36 chains and 144 texts. Six of the twelve alternate chains, barred across founder and word-limit constraints, will follow Human-AI-Human-AI and six will follow AI-Human-AI-Human. As a result, there are 72 human and 72 AI turns in the design. A simulation-based power analysis will employ pilot estimations; chain replication will be raised prior to recruitment if the planned replication is insufficient for the smallest preregistered effect of interest.

4.1 Design

TABLE I CONFIRMATORY DESIGN

Design element	Specification
Transmission lineage	Human-only; AI-only; alternating human-AI
Information bottleneck	Maximum 500 words or 220 words at every generation
Founder material	Two matched 700-word synthetic civilizations
Replication	Three independent chains per cell; four generations per chain
Planned corpus	36 chains; 144 texts; 72 human turns and 72 AI turns

4.2 Founder Cultures and Codebook

Each 700-word founding text will be structured as a recorded cultural specification with 36 atomic facts: 12 random canaries, 12 causal-institutional norms, and 12 sensory-social details. Additionally, each culture will have four nonce labels, three contradiction pairings, and six cross-domain dependencies. Although the surface themes of the two cultures will differ, they will be comparable in terms of length, readability, category counts, dependency density, and narrative relevance. To ensure that category effects cannot be reduced to just one category being more recalled, a different development sample will score each fact's perceived relevance. Until collection is complete, full founder texts will be kept confidential. Cold-start probes will check whether the chosen model can identify any founder material, including nonce labels, prior to collection.

4.3 Participants and Procedure

Seventy-two English-speaking adults will complete one human turn. Participants will only see the text immediately ahead of them after being randomly assigned to a chain position; condition and source metadata will be removed. The source will be revealed during a 12-minute rewrite in order to measure selection and re-expression instead of merely memory. Participants will get the same instructions, telling them not to use artificial intelligence (AI) or other sources and to submit the information as accurately and completely as possible within the provided time frame.

AI turns will use a single, consistent API model and version in order to preserve internal validity. Each round will begin in a fresh environment with no browsing, memory, or access to earlier generations. The human instruction will be preserved exactly, except for the minimal interface language required by the API. The model identification, system prompt, temperature, top-p,

seed when available, token limit, timestamp, and raw output will all be logged. Phrasing, errors, and inconsistencies will not be standardized; only the file format of human and AI outputs will.

4.4 Measures and Coding

Two coders blind to lineage, word-limit condition, founder, and generation will independently code every text. For each registered fact, the codebook will distinguish preserved, mutated, and omitted states. An innovation is a specific claim or named entity with no traceable ancestor in the preceding text or founder specification. Cross-domain dependencies will be scored only when both component facts and their stated relation survive. Each contradiction pair will be classified as preserved, partially lost, explicitly reconciled, or transformed into a different inconsistency. Cohen's kappa will summarize categorical agreement, and prevalence-adjusted agreement will be reported where category imbalance makes kappa unstable. Disagreements will be adjudicated without revealing condition labels.

Fact-level fidelity, mutation rate, innovations per 100 words, regularization of contradictions, and preservation of cross-domain interdependence are the primary results. Compression ratio, lexical variety, pairwise endpoint convergence, semantic distance from the founder, and blind lineage classification are secondary results. Instead of being used in place of registered human coding, automated similarity metrics will be considered supplementary descriptors.

4.5 Statistical Analysis

Fact preservation will be analyzed with a mixed-effects logistic model that includes lineage, word limit, generation, fact type, founder, and preregistered interactions will be used to investigate fact preservation. Repeated observations will be taken into account by random intercepts for chain and atomic fact nested inside founder; if model stability is maintained, a random generation slope by chain will be kept. A negative-binomial mixed model with word count as an offset will be used for innovation counts. A multinomial mixed model will be used to assess contradiction outcomes, whilst a logistic mixed model will be used to investigate dependence survival. Lineages within each generation and bottleneck will be compared in planned contrasts. A permutation test will be used to compare the accuracy of blind categorization to the one-third chance rate. Holm-adjusted p-values, diagnostics, 95% confidence intervals, and effect sizes will be shown. Confirmatory contrasts, missing-output handling, and exclusions

4.6 Exploratory Recombination Extension

A separate exploratory extension will choose one matching human-only and AI-only endpoint from each founder-by-bottleneck cell in order to preserve the recombination question without compromising the confirmatory design. The identical pair will be reconciled into a single coherent culture via a clean AI session and a fresh human transmitter, generating eight more texts. The parent that provides preserved facts, labels, contradictions, and dependencies will be estimated using feature-inheritance scores. These analyzes will be classified as exploratory and won't change the error-control strategy or confirmatory hypotheses.

5 Ethics Feasibility and Limitations

Data collection won't start until the relevant ethical authority has reviewed it. Only consenting individuals may participate in this low-risk activity, which is restricted to a neutral rewriting job. Responses will be saved under anonymous codes and reported collectively; no sensitive personal information will be asked. Stereotyping or misrepresenting a group is less likely when synthetic cultures are used instead of authentic ones. Standard survey, API, and statistical tools are needed for the study. Materials, piloting, power simulation, and preregistration take up two weeks of the eight-week program; collecting takes up two weeks; blinded coding takes up two weeks; and analysis and reporting takes up two weeks.

Effects for one language, one model version, two founding cultures, written transmission, and four generations are all estimated by the design. It cannot prove that every model or actual cultural context behaves in the same way. The bottleneck is controlled rather than kept constant since word restrictions may also lead to convergence regardless of the kind of transmitter. When feasible, model modifications will be avoided during collection; otherwise, they will be recorded as a protocol violation. Generalizability would be tested in a subsequent replication across models, languages, and longer chains.

6 Expected Contribution

The study will give an empirical map of what AI systems, humans, and mixed lineages preferentially preserve, alter, and produce under recurrent transmission. Its main methodological contribution is a controlled, resettable cultural system with complete ground truth and embedded assessments of faithfulness, structure, contradiction, and recombination. The synthetic materials, prompts, codebook, anonymized outputs, preregistration, and analytic code will be made public following data

collection when authorization and competition laws permit. This will enable other researchers to use other models, repeat the experiment, and find out if transmission fingerprints remain after the initial study.

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